simultaneity, 218n27, 222n62
slowing of time, 9–13, 37–40, 76, 194, 197, 198
Smolin, Lee, 219n34
Sophocles: *Oedipus Rex*, 5
spacetime
  and Aristotle, 63–79, 97
  curved, 77, 210
  "empty space," 70–71
  and eternalism, 108–9. *See also* block universe
  fluctuations, 88–90
  granularity, 83–85, 124–25
  gravitational field. *See* gravitational field
  loss of, 95–128, 195
  and quantum mechanics. *See* quantum mechanics
  spinfoam and "quantum spacetime," 126. *See also* loop theory/loop quantum gravity
  structure of, 48–56, 73–79, 106–7, 109, 184, 197–98
  "true" time and Newton's space, 65–79
special relativity theory, 48, 218n26
speed, 145
  acceleration, 17n, 216n7, 216–17n12
  time slowed by, 37–40, 194, 197
spin networks, 126, 127, 224–25n75
spinfoam, 126
Strasburg Cathedral, 59
Strauss, Richard: *Der Rosenkavalier*, 191–92
string theory, 82
suffering, 115, 190–91, 206
sundials, 59–61
synapses, 107–8

Tempier, Étienne, 182
temporal precedence, 48
thermal time, 134–38, 139–40, 142, 196, 222n87
thermodynamics
  second law of, 21–28, 149, 160, 165
  and thermal time, 137, 142. *See also* thermal time
  variables, 226n93
time
  absence, and equations without a time variable, 96, 117–28, 195, 199, 220n47
  Anaximander and the order of, 14–15, 100–101
  arrow of, 19–36, 147–50, 194, 195, 196
  Augustine on, 180–83
  clock-regulated, 59–63. *See also* clocks
  closed lines, 53n
  and diurnal rhythms, 62–63
  dividing and measuring, 58–64
  Einstein on illusion of, 108–9, 114
  and emotion, 141, 200–201, 211–12
  and eternalism, 108–9. *See also* block universe
  flow of. *See* flow of time
  and grammar. *See* grammar
  granularity of, 85–86
  and the gravitational field, 74–79, 90, 124, 195,
  197, 198–99. *See also* gravitational field